

What is the typical histological feature of a pilocytic astrocytoma?

Presentation and Prognosis

Like the symptoms of most mass-occupying lesions, the symptoms of pilocytic astrocytomas are produced by the tumor location. Given its predilection for the cerebellum, common symptoms of an astrocytoma may include gait difficulty, ataxia, dizziness, and nystagmus. A child with new-onset clumsiness is a common presentation.

The prognosis in adults is slightly less favorable than in children, but the 5-, 10-, and 20-year survival rates are 85%, 80%, and 75%, respectively. In children, the 5-year progression-free survival rate after surgical resection is 90%-95%.

What Are Infiltrating Low-Grade Astrocytomas?

INSTRUCTOR NOTE

Again, sometimes the term Diffuse astrocytoma is used.

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Infiltrating low-grade gliomas are WHO grade II tumors. These gliomas are often astrocytic in nature but may also arise from oligodendrocytes.

Incidence and Location

These tumors make up about 10%-15% of gliomas in the adult population and have an incidence of 1.4 cases per 1 million people per year. The median diagnosis age is 30-40 years with a male-to-female ratio of 1:5. In adults, the majority of these astrocytomas occur in the supratentorial region, preferentially in the frontal and temporal lobes.

What is the typical presentation (age, sex, location) for a patient with infiltrating low-grade astrocytoma?

Histology

The definition for low-grade gliomas is hypercellular glial lesions with

QUIZ

 Tap image for quiz

Figure 2

What is the histology of an anaplastic astrocytoma?

Presentation

Presentation largely depends on the exact location of the tumor, but most patients debut with seizures. However, many symptoms also stem from

increased intracranial pressure produced as the tumor grows. Typical signs include headaches, focal deficits, and altered mental status. Tumors in the frontal lobe can cause memory problems, personality changes, and paralysis of the contralateral extremities. Tumors in the temporal lobes can cause seizure and speech and memory problems.

Prognosis

The treatment option of choice is surgical debulking of as much of the tumor as safely possible. If surgery is not possible, radiation alone is a treatment option. For adult patients, chemotherapy can also be used, but there is no approved chemotherapy regimen for children. The 5-year survival rate is about 23.6%. Elderly patients are more likely to die in the first year after diagnosis compared with young adults. The typical median survival is 2-3 years.

What Are Glioblastomas?

Glioblastomas (also known as glioblastoma multiforme [GBM]) are WHO grade IV astrocytomas. Of all glial tumors, these are the most invasive; they grow rapidly and often invade surrounding tissues. These tumors are often devastating and rapidly fatal.

Incidence and Location

The incidence is 2-3 per 100,000 cases per year. These tumors account for more than half (52%) of all malignant primary brain tumors and 17% of both primary and metastatic brain tumors. Glioblastomas usually arise from hemispheric white matter typically in the frontal and temporal lobes. Sometimes they cross the midline via the corpus callosum or anterior

by this bad news, and you step out to give them some privacy to process what they've learned.

Summary

- Gliomas are tumors arising from glial cells, including astrocytes, oligodendrocytes, and ependymal cells.
- Pilocytic astrocytomas, the most common glial tumor in children, are typically benign; they exhibit Rosenthal fibers on histology.
- Infiltrating astrocytomas are WHO grade II tumors that more commonly occur in adults and invade surrounding tissue; on histology, they show hypercellularity with possible areas of microcytosis and calcifications.
- Oligodendrogliomas are tumors of oligodendrocytes that ordinarily myelinate axons in the CNS; oligodendrogliomas have the appearance of fried-egg cells and chicken-wire capillaries on histology.
- Ependymomas tend to occur in the floor of the fourth ventricle in children, causing an obstructive hydrocephalus; in adults, ependymomas usually occur in the spinal cord, creating compressive symptoms.
- Ependymomas have perivascular pseudorosettes, ependymal rosettes, and basal bodies on histology.

Review Questions

1. A 7-year-old previously healthy boy is brought to the emergency department by paramedics after his friend saw him having a seizure. Head computed tomography (CT) showed a mass in the left temporal